

OPERATING SYSTEMS (CS304)

System Calls Overview

Learning Objectives

Understand operating system services, system calls, their types, usage, and importance in process and resource management.

Introduction

System calls provide an interface between user programs and the operating system. They allow applications to request services from the kernel safely and efficiently.

What is a System Call?

A system call is a mechanism through which a user program requests a service from the operating system kernel. Examples include file operations, process creation, memory allocation, and device communication.

Need for System Calls

- Access hardware resources
- Create and manage processes
- Perform file operations
- Communicate with devices
- Ensure system security

Working of a System Call

A user program generates a request, switches to kernel mode, the operating system executes the requested service, and returns control to the user process.

Types of System Calls

1. Process Control
2. File Management
3. Device Management
4. Information Maintenance
5. Communication
6. Protection and Security

Process Control System Calls

Used to create, execute, terminate, and synchronize processes. Examples: `fork()`, `exec()`, `wait()`, `exit()`.

File Management System Calls

Used for file creation, reading, writing, opening, closing, and deletion. Examples: `open()`, `read()`, `write()`, `close()`.

Device Management System Calls

Used to request and release devices, perform input/output operations, and manage device attributes.

Information Maintenance System Calls

Used to obtain system information such as date, time, process IDs, and resource utilization statistics.

Communication System Calls

Used for inter-process communication (IPC) through pipes, shared memory, message queues, and sockets.

Protection System Calls

Used to manage permissions, user authentication, and access control mechanisms.

Advantages of System Calls

- Controlled access to resources
- Improved security
- Efficient resource management
- Hardware abstraction
- Process coordination

Examples in UNIX/Linux

fork() - Create a new process
exec() - Execute a program
wait() - Wait for process completion
read() - Read from file
write() - Write to file

Exam Important Questions

1. Define System Call.
2. Explain the working of a System Call.
3. Discuss different types of System Calls.
4. Explain Process Control System Calls.
5. Write short notes on UNIX System Calls.

Summary

System calls act as a bridge between user applications and the operating system. They provide controlled access to system resources and play a critical role in process, file, and device management.